Ellilpses

Ellipses

Summary

1. **If you're selecting items from a population, draw an ellipse.** The ellipse will represent the tradeoffs between two goals.
   - E.g., if you're admitting students, trading-off between test-scores and diversity.
   - E.g., if you're choosing stories to show to a user, trading-off engagement and quality.
2. **The shape of the ellipse will be determined by the variance and covariance of the attributes.**
   - Implication: if two attributes are highly correlated, the ellipse will be sharp, and there will be very little meaningful tradeoff.
3. **The loss from using a proxy will be proportional to the *excess* covariance.**
   - E.g., if you're using income to proxy for race, the loss will be proportional to the excess loss.

Setup

Observations.

- *Efficiency of a proxy depends on (a) correlation with unobserved outcome, (b) lack of correlation with observed outcome.* Suppose we're trading-off `race` and `score`, we have some Pareto curve (more diversity means sacrificing score). Then suppose we can't use `race` anymore but have to proxy it, e.g. with `income`, we should expect the Pareto curve to shift in, the tradeoff gets worse. Effectiveness of proxy increasing in correlation with race, but decreasing in correlation with score. E.g., suppose income and score had 100% correlation, then . // How much it gets worse depends on (i) correlation with excluded measure; (ii) lack of correlation with other measure.

- *Can extend the Pareto frontier with non-monotonic admission.* Observations: (1) stochastic admission is never optimal, if everything is continuous, you can always replicate with deterministic; (2) just draw the

line E[x_2|x_1], representing the set of achievable points, and the Pareto frontier will be average of a set of those points. If multivariate normal then it's linear.

Beautiful Evidence

Kenneth suggestion on n>2

TLDR: just transform down into n=2 space, then use the same calculation.

> Reparametrize and compute the joint Gaussian for $(x_1, \sum(w_i * x_i), x_2, ...)$
>
> Gram-Schmidt to make them all orthogonal (in the sense of covariance), and now we are considering the joint distribution of (x_1, z, ...), where z is a linear combination of x_1 and $\sum(w_i * x_i)$
>
> conditioning should remove the other components, and we should be able to see that the expectation you want is $E[x_1|ax_1 + bz > k]$, and you can probably reuse your bivariate calculations here for this.
>
> Since Gram-Schmidt is sequential, we don't actually need to worry about the "..." part either, and so just figuring out the joint bivariate distribution of $(x_1, \sum(w_i * x_i))$ and using Gram-Schmidt on it is good enough.

Plus ellipse model says variance and correlation of probabilities, in a ranking case.

- **If just counting VPVs, then efficiency depends only on precision and replacement-rate.**
- **Determinants of replacement-rate.** There are a few different models: (1) suppose people consume a fixed number of VPVs, then replacement rate is determined by the fraction of sessions that are inventory-constrained; (2) suppose that propensity to leave-feed depends on the quality of the story they see, then the replacement rate will depend on the quality of the removed stories (if they are low-quality, could *increase* feed time); (3) suppose that propensity to return to FB depends on the quality of stories they're seeing, then long-run replacement rate will again depend on the quality of the removed stories.

3. **When two outcomes are positively correlated then tradeoffs will be relatively inefficient.** E.g. suppose p_A and p_B are positively correlated, then a given reduction in `A` will cost a lot of `B`. If instead they were negatively correlated then we could reduce misinformation a lot with a low MSI cost, i.e. tradeoffs would be relatively efficient.
   - The relevant correlation is *within-user*, not *between-user*. This also explains the effect of inventory: if p_B tends to be higher for inventory-

constrained people, it implies a high within-user correlation between p_B and p_A (because $p_A = p_B = 0$ when the user sees nothing), and so implies relatively low efficiency in the tradeoff.

4. **When there is low variance in p_B then tradeoffs against B will be relatively efficient.** E.g. suppose that all content had roughly the same value of p_B, then we could make large changes to A with relatively little impact on B.
5. **Improved quality of predictions will increase the efficiency of tradeoffs.** An improvement to the accuracy of $p(A)$ will shift outwards the set of feasible combinations of `A` & `B`, and will improve the average efficiency of all tradeoffs. Even a model with relatively low precision can have high efficiency if, on the margin, the variance of p_B is low. This would be the case if, for example, our model was able to identify small variations in p_A among stories with similar levels of p_B, allowing us to increase A at a small cost in B.
6. **Network effects make tradeoffs more favorable to the outcome with larger network effects.** We can distinguish between the *user-level* tradeoff, which we measure in experiments, and the *aggregate* tradeoff, which includes network effects. If both `A` and `B` have equally large network multipliers, then the user-level and aggregate tradeoff curves will look the same. If instead `A` has a larger network effect, then the aggregate tradeoffs will be relatively more favorable to `A` than those measured in user-level experiments.

Application to Misinformation

We are interested in how our misinformation prediction models (and other ranking intervensions) trade-off between the prevalence of misinformation and other goals, e.g. total engagement (could be defined as MSI, WYT, time-spent, etc.).

Graphically:

1. **The slope of the frontier**, at production, will be equal to the weight on `P(misinfo)` in our value model. This would be true in a simplified setup, it's probably not quite true in practice for various other reasons, but I think it's a resonable starting-point (and we can test whether it's true).
2. **The average efficiency of the tradeoff depends on a number of factors:**
   - **The correlation between `p(engagement)` and `p(misinformation)`.** –If they are positively correlated the tradeoff will be inefficient (curve will be sharp), if they are negatively correlated the tradeoff will be efficient (curve will be flat).
   - **The incrementality of misinformation.** – If most misinformation is consumed by inventory-constrained people, then the tradeoff will

be relatively inefficient (curve will be sharp) .

- **The variance in `p(misinformation)`.** – If it has high variance – i.e. it's a better prediction model – then the tradeoff will be relatively efficient (curve will be flat).
- **The network effects of misinformation.** – If misinformation has higher network effects than non-misinformation (which seems likely), this will make the aggregate tradeoff more efficient than what is measured in user-level experiments.

3. **To judge the effectiveness of a misinformation demotion, we should look at the *average efficiency*.** The average efficiency of a demotion could be high even when the model has low precision or recall.
   - Intuitively, a low-precision classifer can achieve significant efficiencies if it rearranges stories with similar $p(engagement)$, but with differences in $p(misinformation)$.
   - We can calculate the average efficiency of our demotion by looking at a holdout, and making adjustments for network effects.
   - **Key determinants of tradeoff:**

     $$\frac{\Delta\text{MSI}}{\Delta\text{misinfo}} = \frac{\text{(MSI of labelled-misinfo)-(MSI of replacement content)}}{\text{precision of labels}}.$$

     All of these values should be interpreted as *marginal* values: marginal precision, and MSI of marginal misinfo (technically the MSI of *labelled* misinfo), and marginal replacement content.
     E.g., suppose we want to reduce misinfo from 0.5% to 0.25%, and the following are true:
     - marginal content generates 1/4 of the MSI of the average content.
     - misinfo has about average MSI
     - precision is 10%

     Then the effect on MSI will be around 2% (=0.25%(1-0.25)/0.1).

Model

Setup

Suppose we're choosing N out of M items from a set. Each item is defined by a set of characteristics $(x_1, \ldots, x_n)$, with some joint distribution. We can define a *value function* as a weighted average of characteristics,

$$v = w_1 x_1 + \ldots + w_n x_n. \quad \text{(value function)}$$

The threshold value v^* which admits N of M will satisfy:

$$P(\sum_{i=1}^{n} w_i p_i > v^*) = \frac{N}{M}.$$

We can define the average outcomes among those selected,

$$X_i = E[p_i | \sum_i w_i p_i > w^*].$$

And define an objective function over those average outcomes:

$$U(X_1, \ldots, X_n). \text{ (objective function)}$$

We can illustrate this in two dimensions:

The left-hand figure show the joint distribution of the two characteristics (x_1 and x_2), the black curve in the right-hand figure shows the achievable combinations of *average* characteristics, i.e. X_1 and X_2. For completeness the dashed grey ellipse shows every combination, including negative coefficients ($w_1, w_2 < 0$), and any point inside the ellipse is also achievable by some other admission rule.

Solution with Two Gaussian Variables

When we have two outcomes, x_1, x_2 and they are normally distributed, with mean zero, variances σ_1, σ_2, and correlation ρ, then we can directly solve for X_1 and X_2 as a function of the relative weights ($w = \frac{w_2}{w_1}$)::

$$\begin{aligned}
X_1 =& \frac{\sigma_1^2 + w\rho\sigma_1\sigma_2}{\sqrt{\sigma_1^2 + w^2\sigma_2^2 + 2\rho w\sigma_1\sigma_2}}\gamma \\
X_2 =& \frac{w\sigma_2^2 + \rho\sigma_1\sigma_2}{\sqrt{\sigma_1^2 + w^2\sigma_2^2 + 2\rho w\sigma_1\sigma_2}}\gamma \\
\gamma =& \frac{\phi(\frac{v^*}{var(v)})}{\Phi(\frac{v^*}{var(v)})} P(v \geq v^*).
\end{aligned}$$

We can also get a direct expression for the shape of the Pareto frontier, it will be an ellipse with the same shape as the joint distribution of x_1 and x_2:

$$X_1 = \sigma_1 \sqrt{(1-\rho^2)(1 - \frac{X_j^2}{\sigma_2^2})} + \rho\frac{\sigma_1}{\sigma_2} X_j.$$

Observations

The optimal value function is a weighted-average

Any point on the Pareto frontier can be achieved by a linear value function (i.e., a weighted average). If your objective function is itself linear, then you can use your objective-function weights in the value-function (e.g., if X_2 is worth twice as much as X_1 in your objective function, then just set $\frac{w_2}{w_1} = 1$).

The effect of a marginal change in weight is the weight itself

In a two-outcome model we can write:

$$\frac{dX_2/dw_1}{dX_1/dw_1} = -\frac{w_2}{w_1}.$$

E.g., suppose currently outcome #2 has twice the weight as outcome #1 ($\frac{w_2}{w_1} = 2$), then any small change in weights which increases X_1 by δ will decrease x_2 by $\frac{1}{2}\delta$.

The effect of increasing admission

The relative effect of admitting more is equal to the relative weights. If you increase the number admitted (increase N), then the relative increase in the two outcomes will be equal to the weights:

$$\frac{dX_2/dN}{dX_1/dN} = \frac{dX_2/dv^*}{dX_1/dv^*} = \frac{w_2}{w_1}.$$

E.g., if you're increasing admission by 10%, and outcome #2 has twice the weight of outcome #1 ($\frac{w_2}{w_1} = 2$), then the increase in outcome #2 will be twice the increase in outcome #1.

1. **The curvature of the tradeoff depends on covariance.** I.e., what is:

   $$\frac{dX_2^2/d^2w_1}{dX_1^2/d^2w_1} = ?.$$

   Broadly speaking, we should expect low curvature if there is negative correlation between p_i and p_j, but high curvature if there is positive correlation.

2. **What is the overall efficiency of a term?** Sometimes we don't just want to know the *marginal* tradeoff, we also want to know the *average* tradeoff. Suppose we have a given $\bar{w}_2$

   $$\frac{X_2(w_2 = \bar{w}_2) - X_2(w_2 = 0)}{X_1(w_2 = \bar{w}_2) - X_1(w_2 = 0)} = ?.$$

   Broadly speaking this again should depend on the correlation between p_i and p_j.

3. **What is the impact of an improvement in detection.** Suppose we improve our prediction model, p_j. In a simple case we could think of this as increasing the variance of p_j, i.e. σ_j, and so we could ask these questions:

   $$\frac{dX_j}{d\sigma_j} = ?, \quad \frac{dX_i}{d\sigma_j} = ?$$

Observations

1. **As variance increases then threshold increases.** Intuitively, as predictions get better, then we have a higher threshold (assuming that we are currently showing a minority of the content). I.e., w^* is increasing in σ_1, σ_2, and ρ, as long as $w^* > \mu$.

2. **Suppose that we only care about outcome i,** i.e. set $v_j = 0$, then w is perfectly correlated with p_1, and the expression reduces down to simply σ_i.

3. **Suppose perfect correlation between the predictions.** I.e., assume $\rho = 1$, then we have:
$$X_i = \sigma_i \gamma, \quad X_j = \sigma_j \gamma.$$

4. **Suppose no correlation between the predictions.** I.e., assume $\rho = 0$, then we have:
$$\begin{aligned} X_i =& \frac{\sigma_i^2}{\sqrt{\sigma_i^2 + v^2\sigma_j^2}}\gamma \\ X_j =& \frac{v\sigma_j^2}{\sqrt{\sigma_i^2 + v^2\sigma_j^2}}\gamma \end{aligned}$$

   Thus each is decreasing in v and increasing in σ_i. Derivatives with respect to weights:
$$\begin{aligned} \frac{dX_i}{dv} =& -v\sigma_j^2\sigma_i^2(\sigma_i^2 + v^2\sigma_j^2)^{-3/2}\gamma < 0 \\ \frac{dX_j}{dv} =& -v^2\sigma_j^4(\sigma_i^2 + v^2\sigma_j^2)^{-3/2}\gamma + \sigma_j^2(\sigma_i^2 + v^2\sigma_j^2)^{-1/2}\gamma > 0 \\ \frac{dX_j/dv}{dX_i/dv} =& \frac{-v^2\sigma_j^4 + \sigma_j^2(\sigma_i^2 + v^2\sigma_j^2)}{-v\sigma_j^2\sigma_i^2} \\ =& \frac{-v^2\sigma_j^2 + \sigma_i^2 + v^2\sigma_j^2}{-v\sigma_i^2} = -\frac{1}{v}. \end{aligned}$$

   Derivatives with respect to noise, we can immediately see the signs:
$$\frac{dX_i}{d\sigma_i} > 0, \quad \frac{dX_i}{d\sigma_j} < 0$$

5. **Effect of improvement in prediction models.** You can think of the effect of improving a prediction model as increasing variance, i.e. $\sigma_i \uparrow$. I'm not sure whether this can *decrease* the expected number of actions of this type. Intuitively can happen outside the Gaussian case: e.g. suppose that most of the probability mass is above the threshold, then increasing the variance will tend to mainly push things below the threshold.

Inventory consumption

As inventory consumption increases, the frontier will shift out, but less than proportionally with the increase in consumption.

Variance

- If greater variance in both metrics, then the range of tradeoffs will be broader.

Suppose we increase the variance of p_A, e.g. by improvements to the prediction model. This shifts out the Pareto frontier in the direction of A.

Improve predictions of A -> increase the variance of p_A -> shifts out Pareto frontier -> increase A and *decrease* B, but this is efficient. (May not hold if there is positive correlation in predictions).

Average efficiency

[not sure if we get a closed-form expression for this.]

Proofs

We want to show that with a joint-normal distribution of (x_1, x_2), the Pareto frontier for admitting N items will be elliptical and exactly corresponding to an isovalue of the distribution.

From here on the notation differs a little from the rest of the note, I haven't updated it yet. We start with this:

$$w = p_i + vp_j$$

We can write the covariance matrix as:

$$Cov \begin{bmatrix} p_i \\ p_j \\ w \end{bmatrix} = \begin{bmatrix} \sigma_i^2 & \sigma_i\sigma_j\rho & \sigma_i^2 + v_j\rho\sigma_i\sigma_j \\ \sigma_i\sigma_j\rho & \sigma_j^2 & \rho\sigma_i\sigma_j + v_j\sigma_j^2 \\ \sigma_i^2 + v_j\rho\sigma_i\sigma_j & \rho\sigma_i\sigma_j + v_j\sigma_j^2 & \sigma_i^2 + v_j^2\sigma_j^2 + 2\rho v_j\sigma_i\sigma_j \end{bmatrix}$$

Where ρ is the correlation coefficient between p_i and p_j. We wish to know the total number of actions of type i, X_i, produced for a given threshold w^* on total weight:

$$X_i = P(w \geq w^*)E[p_i | w \geq w^*].$$

First we can calculate the conditional expectations:

$$
\begin{aligned}
E[p_i|w \geq w^*] =& \sigma_i \frac{Cov(p_i, w)}{\sqrt{Var(p_i)Var(w)}} \frac{\phi(\frac{w^*}{\sqrt{Var(w)}})}{\Phi(\frac{w^*}{\sqrt{Var(w)}})} \\
=& \sigma_i \frac{\sigma_i^2 + v_j \rho \sigma_i \sigma_j}{\sqrt{\sigma_i^2(\sigma_i^2 + v_j^2 \sigma_j^2 + 2\rho v_j \sigma_i \sigma_j)}} \frac{\phi(\frac{w^*}{\sqrt{Var(w)}})}{\Phi(\frac{w^*}{\sqrt{Var(w)}})} \\
=& \frac{\sigma_i^2 + v_j \rho \sigma_i \sigma_j}{\sqrt{\sigma_i^2 + v_j^2 \sigma_j^2 + 2\rho v_j \sigma_i \sigma_j}} \frac{\phi(\frac{w^*}{\sqrt{Var(w)}})}{\Phi(\frac{w^*}{\sqrt{Var(w)}})}
\end{aligned}
$$

Now let us suppose that the number of stories consumed is fixed, then $P(w \geq w^*)$ is a constant, and $\frac{w^*}{\sqrt{Var(w)}}$ will be a constant, and we will define:

$$
\begin{aligned}
\gamma =& \frac{\phi(\frac{w^*}{\sqrt{Var(w)}})}{\Phi(\frac{w^*}{\sqrt{Var(w)}})} P(w \geq w^*) \\
X_i =& \frac{\sigma_i^2 + v \rho \sigma_i \sigma_j}{\sqrt{\sigma_i^2 + v^2 \sigma_j^2 + 2\rho v \sigma_i \sigma_j}} \gamma \\
X_j =& \frac{v \sigma_j^2 + \rho \sigma_i \sigma_j}{\sqrt{\sigma_i^2 + v^2 \sigma_j^2 + 2\rho v \sigma_i \sigma_j}} \gamma
\end{aligned}
$$

We start with the expressions for X_i and X_j, as derived above. Then I fed them into Mathematica, to solve:

```
Fi[v_,p_,si_,sj_]:= (si^2+v p si sj)/Sqrt[si^2+v^2sj^2+2v p si sj]
Fj[v_,p_,si_,sj_]:=(v sj^2+p si sj)/Sqrt[si^2+v^2sj^2+2v p si sj]
Solve[{Xi==Fi[v,p,si,sj],Xj==Fj[v,p,si,sj]}, {Xi,v}]

{Xi -> (sj^4*Sqrt[-(((-1 + p^2)*si^2*sj^2)/(sj^2 - Xj^2))] -
     sj^2*Xj^2*Sqrt[-(((-1 + p^2)*si^2*sj^2)/(sj^2 - Xj^2))] +
     p*si*sj*Xj*(sj^2 - Xj^2) + Xj*Sqrt[(-1 + p^2)*si^2*sj^2*Xj^2*
        (-sj^2 + Xj^2)])/(sj^4 - sj^2*Xj^2),
  v -> (p*si*sj*(-sj^2 + Xj^2) - Sqrt[(-1 + p^2)*si^2*sj^2*
       ((-sj^2)*Xj^2 + Xj^4)])/(sj^4 - sj^2*Xj^2)}
```

Notebook @ https://www.wolframcloud.com/env/tomcunningham/Tradeoffs%20v2.nb

Mathematica gives a big ugly solution but it can be simplified a lot (here $s = \sigma$ and $p = \rho$):

$$
\begin{aligned}
X_i =& \frac{(s_j^4 - s_j^2 X_j^2)\sqrt{\frac{-(-1+p^2)*si^2*sj^2}{sj^2-Xj^2}} + ps_i s_j X_j(s_j^2 - X_j^2) + X_j\sqrt{(-1+p^2)s_i^2 s_j^2 X_j^2(-s_j^2 + X_j^2)})}{s_j^4 - s_j^2 X_j^2} \\
=& \frac{(s_j^4 - s_j^2 X_j^2)s_i s_j\sqrt{\frac{(1-p^2)}{s_j^2-X_j^2}} + ps_i s_j X_j(s_j^2 - X_j^2) + X_j^2 s_i s_j\sqrt{(1-p^2)(X_j^2 - s_j^2)})}{s_j^2(s_j^2 - X_j^2)} \\
=& \frac{-s_i s_j^3\sqrt{(1-p^2)(X_j^2 - s_j^2)} + ps_i s_j X_j(s_j^2 - X_j^2) + X_j^2 s_i s_j\sqrt{(1-p^2)(X_j^2 - s_j^2)})}{s_j^2(s_j^2 - X_j^2)} \\
=& \frac{s_i s_j(X_j^2 - s_j^2)\sqrt{(1-p^2)(X_j^2 - s_j^2)} + ps_i s_j X_j(s_j^2 - X_j^2)}{s_j^2(s_j^2 - X_j^2)} \\
=& \frac{-s_i s_j\sqrt{(1-p^2)(X_j^2 - s_j^2)} + ps_i s_j X_j}{s_j^2} \\
=& s_i\sqrt{(1-p^2)(1 - \frac{X_j^2}{s_j^2})} + p\frac{s_i}{s_j}X_j \\
=& X_j\rho\frac{\sigma_i}{\sigma_j} \pm \sqrt{1 - \rho^2 - (1-\rho^2)\frac{X_j^2}{\sigma_j^2}}
\end{aligned}
$$

which is very close to the isovalue below, I think must be equal once we fix the algebra.

Isovalue of a Normal distribution. The general expression for an isovalue for a given density k will be:[1]

$$
k = \frac{x_1^2}{\sigma_1^2} + \frac{x_2^2}{\sigma_2^2} - 2\rho\frac{x_1 x_2}{\sigma_1\sigma_2}.
$$

We can rearrange this to get:

$$
\begin{aligned}
x_2 =& x_1\rho\frac{\sigma_2}{\sigma_1} \pm \frac{\sigma_2}{\sigma_1}\sqrt{-p_1^2 + p_1^2\rho^2 + k\sigma_1^2} \\
=& x_1\rho\frac{\sigma_2}{\sigma_1} \pm \frac{\sigma_2}{\sigma_1}\sqrt{k\sigma_1^2 - (1-\rho)x_1^2}.
\end{aligned}
$$

We can draw this here:

[1] From http://athenasc.com/Bivariate-Normal.pdf , they say "This is Section 4.7 of the 1st edition (2002) of the book Introduction to Probability, by D. P. Bertsekas and J. N. Tsitsiklis."

Relevant Literature

My prior stuff:

- `prussia-log-2020h2` note on randomizing entry. Diagram showing quota vs score cutoff.
- `2-1-trading-off-multiple-objectives`
  - If X and Y are positively correlated,
  - If prediction models get better, increases variance of predictions, can increase or decrease outcome.
- `2019-10-18-tradeoffs in ranking systems`
- Ellison & Pathak (AER, 2021) *"The Efficiency of Race-Neutral Alternatives to Race-Based Affirmative Action: Evidence from Chicago's Exam Schools"*
  - proxying race with SES leads to inefficiency.
  - Core problem that best-performing kids in poor neighborhoods are white. > "The example shows that even a highly correlated proxy can be of no help if it is separately related to the admissions score."
- Chan and Eyster (2003) "possible to increase representation of an under-represented group by adding noise to the decision process."
  - Q: adding noise vs non-monotonic?

Operations research literature on Pareto frontiers

Matthew Daskilewicz (2013) PhD Thesis -> Reduce a Pareto sphere into a simplex (barycentric coordinates), where each vertex represents maximum of just that attribute.

Yusheng Li et al. (2012) *Approximating Pareto curves using the hyper-ellipse*

Yusheng Li et al. (2003) *Minimum Effort Approximation of the Pareto Space of Convex Bi-Criteria Problems*

Not super-relevant: I don't think they're applied to Pareto surface for *selection* problems.

Economics Literature ("abstract selection problem")

Hazel & Lush (1942) *"THE EFFICIENCY OF THREE METHODS OF SELECTION"* - discusses using attribute-specific thresholds vs having a weighted-average.

Hazel & Lush (1943) *"THE GENETIC BASIS FOR CONSTRUCTING SELECTION INDEXEs"*

Gary-bobo and Alain Trannoy (2007) *Efficient Tuition Fees and Examinations* They consider when you're setting two separate thresholds: (1) tuition fees, and

(2) exam-score threshold. A nice observation that it can make sense to use both thresholds in admitting people.